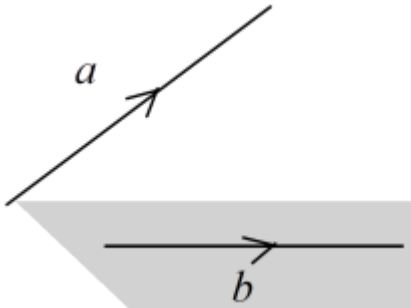
1. Vectors & Transformations

1.1. Vectors

- **Vector quantities** have both magnitude and direction
- Vectors a and b represented by the line segments can be added using the 'parallelogram rule' or 'nose-to-tail method'



- Vectors are also written like $\overrightarrow{AB}$

Scalar Quantities

- **Scalar Quantity:** has a magnitude but no direction

Multiplication by a Scalar

- The negative sign reverses the direction of the vector
- For Eg, $3\,(a\ b) = (3a\ 3b)$

Column Vector

$$(x\ y)$$

- Top number = Horizontal component
- Bottom number = Vertical component

Parallel Vectors

- Vectors are parallel if they have the same direction
- In general, the vector $k\,(a\ b)$ is parallel to $(a\ b)$

Magnitude of a Vector

- In general, if $\vec{x} = (m\ n)$, $|\vec{x}| = \sqrt{(m^2 + n^2)}$

Addition and Subtraction of Vectors

Addition of Vectors

- Add vectors using the tip-to-tail rule.
- The **resultant vector** goes from the start of the first to the end of the last.
- Can also add using the parallelogram method.
- $\vec{AB} + \vec{BC} = \vec{AC}$
- In component form:
  - $(a_x\ a_y) + (b_x\ b_y) = (a_x + b_x\ a_y + b_y)$

Subtraction of Vectors

- To subtract, reverse the vector:

$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$

- $-\vec{B}$ has same length as $\vec{B}$ but opposite direction
- In component form:
  - $(a_x\ a_y) - (b_x\ b_y) = (a_x - b_x\ a_y - b_y)$

Position Vectors

- A position vector locates a point relative to an origin O.
- The position vector of A is written as $\vec{OA}$
- In coordinates:
  - $\vec{OA} = (x\ y)$
- To find a vector between two points:
  - $\vec{AB} = \vec{OB} - \vec{OA}$
- Position vectors can be added or subtracted like regular vectors

Equal Vectors

- Two vectors are equal if they have the same magnitude and direction

Parallel Vectors

- Two vectors are parallel if they have the same/opposite direction, but different magnitudes, and the same ratio of magnitude

Collinear Vectors

- Vectors that lie along the same line or parallel lines. They can be parallel (pointing in the same direction) or anti-parallel (pointing in opposite directions)

Gradient of a Vector

The gradient m of a vector $\vec{v} = (x\ y)$ is given by $m = \frac{y}{x}$.

If the vector represents a line joining two points $A(x_1, y_1)$ and $B(x_2, y_2)$, then $\vec{AB} = (x_2 - x_1\ y_2 - y_1)$ and $m = \frac{y_2 - y_1}{x_2 - x_1}$.

Negative Vectors

- Vectors that have the same magnitude (length) as its corresponding positive vector but points in the opposite direction

1.2. Transformation

Reflection (M)

- Image & object are the same size
- Image and Object are equidistant from the line of reflection
- Lines joining each point and its image are perpendicular to the mirror line

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- Standard Cases of Reflection
  - Reflection in the x-axis

    Rule: $(x, y) \to (x, -y)$
  - Reflection in the y-axis

    Rule: $(x, y) \to (-x, y)$
  - Reflection in the line y = x

    Rule: $(x, y) \to (y, x)$
  - Reflection in the line y = -x

    Rule: $(x, y) \to (-y, -x)$

Worked Example

Q. Transform shape ABC by reflection in the line $y = x$. $A(2, 1), B(4, 1), C(4, 3)$

Step 1: Apply the Rule

- Reflection in $y = x \to$ swap coordinates:

  $(x, y) \to (y, x)$

Step 2: Transform each point

- $A(2, 1) \to A'(1, 2)$
- $B(4, 1) \to B'(1, 4)$
- $C(4, 3) \to C'(3, 4)$

Step 3: Plot the image

- Plot A', B', C' and join them to form the reflected image.

Step 4: Check

- The image should be the same size
- Each point and its image should be the same distance from $y = x$

How to Describe a Transformation (Reflection)

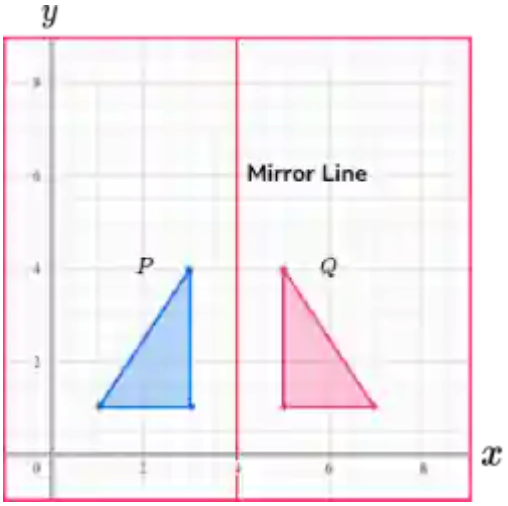
1. Identify the line of reflection

- Look for a line that is exactly between points and their images
- Check equal distances:
  - eg: if A and A' are both 3 units from the x-axis $\to$ line of reflection = x-axis.

2. State the transformation clearly

   Use the phrase:

   "Shape A is mapped onto Shape B by a reflection in the line ____"



Rotation (R)

- The centre, angle, and direction of rotation are needed to describe a rotation
- A clockwise rotation is negative, and an anticlockwise rotation is positive
- Size and shape of rotation do not change
- Standard Cases of Rotation:
  - Rotation 90° anticlockwise about the origin (270° clockwise)

    Rule: $(x, y) \to (-y, x)$
  - Rotation 90° clockwise about the origin (270° anticlockwise)

    Rule: $(x, y) \to (y, -x)$
  - Rotation $180^{\circ} about the origin

    Rule: $(x, y) \to (-x, -y)$

Worked Example

Q. Rotate triangle ABC 90° anticlockwise about the origin. $A(2, 1), B(4, 1), C(4, 3)$

Step 1: Use the rule

- For $(x, y) \to (-y, x)$

Step 2: Transform each point

- $A(2, 1) \to A'(-1, 2)$
- $B(4, 1) \to B'(-1, 4)$
- $C(4, 3) \to C'(-3, 4)$

Step 3: Plot the image

- Plot A',B',C' and join them to form the rotated image

How to find the Centre of Rotation:

- Join the corresponding vertices of the shapes
- Locate their midpoints
- Draw a perpendicular to both the midpoints

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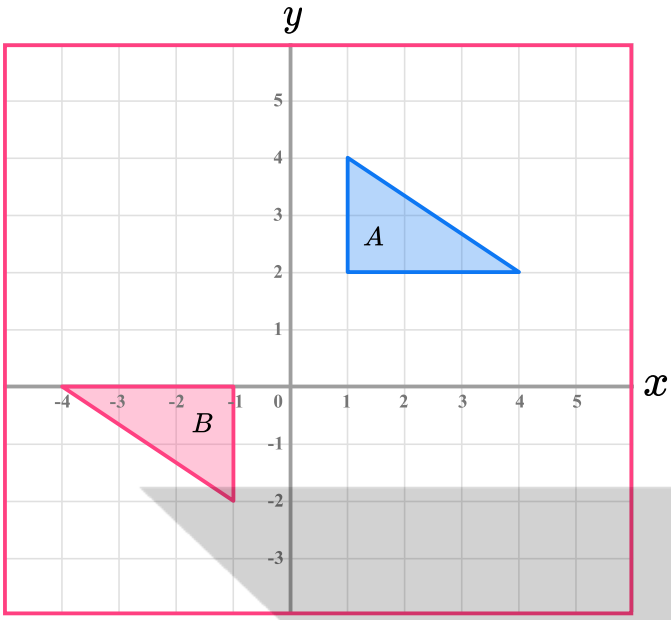
- The intersection is the centre of rotation



- Rotation of 180 degrees about the point (0,1)

Translation (T)

- A translation slides a shape without rotating or reflecting it.
- To describe a translation, you must give the translation vector:

$$\begin{pmatrix} a \\ b \end{pmatrix}$$

- Right is $+x$, left is $-x$, up is $+y$, down is $-y$
- Size, shape, and orientation do not change under translation.
- Standard Form of Translation:
  - A translation by vector:

$$\begin{pmatrix} a \\ b \end{pmatrix}$$

- means:
  - Move a units horizontally
  - Move b units horizontally
- Rule: $(x, y) \to (x + a, y + b)$

Worked Example

Q. Translate triangle ABC by vector $\begin{pmatrix} 5 \\ -2 \end{pmatrix}$ where $A(2,4), B(5,4), C(5,7)$

Step 1: Use the rule

- $(x, y) \to (x + 5, y - 2)$
- For the given vector, we move 5 units right and 2 units down

Step 2: Transform each point

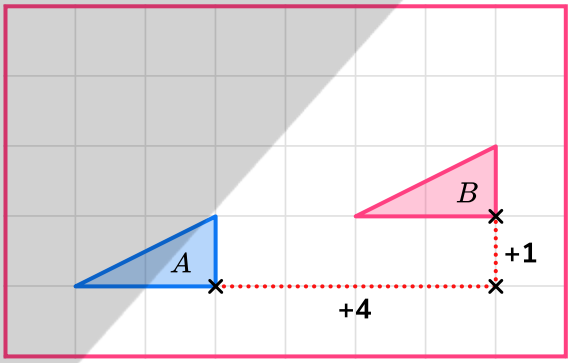
- $A(2,4) \to A'(2+5, 4-2) = A'(7,2)$
- $B(5,4) \to B'(5+5, 4-2) = B'(10,2)$
- $C(5,7) \to C'(5+5, 7-2) = C'(10,5)$

Step 3: Plot the image

- Plot $A'(7,2), B'(10,2), C'(10,5)$ and join them to form the translated triangle

How to find the Translation Vector

- $T = I - O$
- Compute the differences
  - Horizontal Shift: $x_2 - x_1$
  - Vertical Shift: $y_2 - y_1$
- Example:
  - Object: $P(2,3)
  - Image: P'(7,1)
  - $T = \begin{pmatrix} 7-2 \\ 1-3 \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$
  - So the translation is 5 right and 2 down



- Translation by vector $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$

Enlargement (E)

- To describe an enlargement, you must state:
  - scale factor (k)
  - centre of enlargement
- Size of the image changes, but the shape does not change
- Effect of Scale Factor
  - If k>1 → the image becomes larger than object
  - If 0<k<1 → the image becomes smaller than object
  - If k = 1 → the image is unchanged
  - If -1<k<0 → the image becomes smaller than the object, but it is on the opposite side of centre
  - If k<-1 → the image becomes bigger and on the opposite side of the centre.
- Useful Formulae:
  - $\text{Scale Factor} = \frac{\text{Length of Image}}{\text{Length of Object}}$
  - $\text{Area of Image} = k^2 \times \text{Area of Object}$
  - $\text{Image Point} = \text{Centre} + k(\text{Object Point} - \text{Centre})$

Worked Example

Q. Enlarge triangle ABC by scale factor $k = 2$ about the centre $O(0,0)$

A(1,2), B(3,2), C(3,5)

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Step 1: Use the rule

- $(x, y) \rightarrow (2x, 2y)$
- Because scale factor $k = 2$

Step 2: Transform each point

- $A(1, 2) \rightarrow A'(2, 4)$
- $B(3, 2) \rightarrow B'(6, 4)$
- $C(3, 5) \rightarrow C'(6, 10)$

Step 3: Plot the image

- Plot $A'(2, 4), B'(6, 4), C'(6, 10)$ and join to form the enlarged triangle

How to find the Centre of Enlargement

1. Join the corresponding vertices of the object and image
2. Extend the lines backward
3. All lines will intersect at one point
4. That intersection is the centre of enlargement

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